Structuring Source Information in Early Japanese Dictionaries Using TEI/XML and RDF

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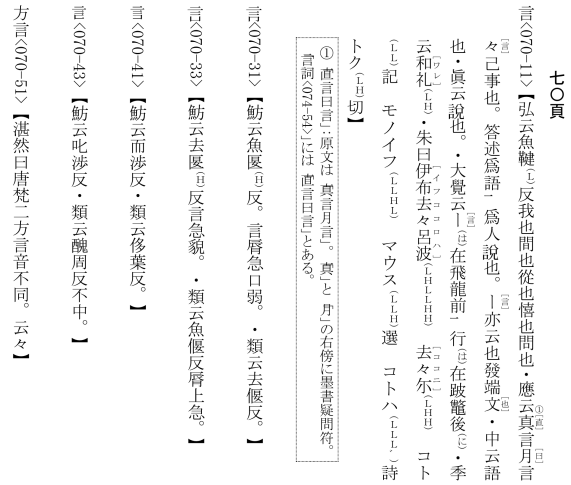
Introduction

The Zushoryō edition of *Ruiju Myōgishō*, compiled by a Buddhist scholar-monk at the end of the 11th century, stands out as an essential resource in the history of Japanese lexicography. This dictionary uniquely integrates extensive references to Buddhist scriptures, classical Chinese texts, and various historical sources. Its intricate annotation system, detailing phonetic notations, semantic explanations, and orthographic variations, exemplifies the intellectual and cultural dynamics of the Heian period.

Textual Features and Encoding Challenges

Ruiju Myōgishō presents substantial challenges for digital text encoding due to its complex, multi-layered annotation structure. The dictionary predominantly includes multi-character entries, unlike previous works focused primarily on single-character entries. Each entry is annotated with detailed citations, which necessitates precise encoding to preserve these complex textual relationships digitally.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RDF facilitates the representation of non-linear, multidirectional relationships, thus overcoming the limitations of traditional XML encoding. By integrating RDF, this research achieves a more accurate and flexible depiction of the dictionary's extensive citation practices, linking multiple textual sources, author attributions, and contextual metadata seamlessly.

Hybrid TEI/XML and RDF Model Implementation

The core contribution of this research lies in developing a hybrid data modeling approach, integrating TEI/XML and RDF. Initially, a TEI/XML schema tailored to *Ruiju Myōgishō*'s specific textual characteristics was created, followed by the application of RDF ontologies to represent citation relationships comprehensively. The resulting hybrid model not only captures the dictionary's complexity but also facilitates interoperability with other digital textual resources, significantly enhancing the potential for cross-textual analyses and collaborative scholarly efforts.

Broader Implications and Future Work

The developed hybrid model has broader implications for historical lexicography and digital humanities. It exemplifies a successful integration of traditional philological research with advanced digital methodologies, promoting a deeper understanding of textual transmission and intellectual history. Future research will aim to further refine RDF ontologies, incorporating additional semantic layers such as interpretative variants and detailed contextual metadata, and to explore linked-data applications that integrate *Ruiju Myōgishō* with other East Asian textual corpora.

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